I. Experimental Evaluation

A. Data, Protocol and Complete Methods

We evaluate closed-set radio-frequency fingerprinting on the ManySig and ManyRx subsets of WiSig [1]. Source and target share the registered transmitter (TX) classes. Training uses dates D1–D3 (March 8, 15 and 23, 2021); D0 (March 1) is the earlier unseen date. Three disjoint populations test a date shift (source RX on D0), a receiver shift (target RX on D1–D3), and both shifts (target RX on D0). The date result is not future-date extrapolation. ManySig has six TX classes; ManyRx retains ten classes and unequal or missing cells. We do not pool their accuracies.

ManySig source TX/RX/day cells contain 70 labeled (L), 630 TX-hidden unlabeled (U), and 300 validation (V) packets, giving 6300/56700/27000 for F0. The default L fraction is 10% of the training pool, with additional validation labels disclosed separately. The 1% and 5% conditions use 630 and 3150 L packets and preserve V. ManyRx has 6533 L, 58828 U and 28023 V packets. Source-count experiments use natural budgets at $m = 1, 2, 3, 4, 6$ and six fixed target RX; F0's five-source/seven-target result is not inserted into that curve. We do not introduce a fixed-L budget or a separate chronological training experiment.

Equalized IQ is centrally cropped to 256 samples and normalized to unit RMS before channel transformation, giving a 2×256 float32 input. Each observation retains its physical packet identity. We test clean and all six practical suburban/urban channel views from the same fixed received-IQ handles. These are synthetic residual-channel stress tests, not measurements of an operational satellite link. DURA processes clean as well as impaired inputs, without scene-aware bypass or additional output renormalization.

We compare the complete RIEI [2], DRIFT-v2 [3], IB-RFF [4] and SDG [5] methods. Their original identity head is trained jointly with the backbone; no post-hoc head is fitted. Each Full recipe is evaluated with channel augmentation and ordinary pseudo-labeling (PL) independently off/on. Cells A/B have channel off and C/D have channel on; A/C have PL off and B/D have PL on. The main comparison fixes D rather than selecting a target-optimal cell. All four cells remain in the supplement; external CE-only variants are excluded.

External channel augmentation uses the actual practical ResidualChannel engine in residual/post-sync/no-equalization mode. Training samples low suburban, mid urban and low urban with probabilities .40/.35/.25, applying them with probabilities .30/.60/.80 over budget intervals $[0, .20)$, $[.20, .45)$ and $[.45, 1)$. The schedule resets within each IB stage. A common .68 auxiliary TX-CE slot is retained in all four cells; unselected or channel-off samples use original L IQ. Native Full losses and internal augmentations remain active. External PL uses the current model at confidence .95 and weight 1, whole-U-batch reduction and no EMA or complex gate. The independently completed external matrix and its source-only early-stopping behavior are reused without retraining.

The selected PAMF-REPL identifier keeps the original supervised CRIC, ordinary/raw paired symmetric KL, two As-label CE passes with As KL zero, label smoothing .01 and six L forwards (768 packet evaluations/update). Its original L As uses the three practical environments uniformly; it is not silently replaced by the external auxiliary-concatenation recipe. The matched Sup and ordinary-PL controls keep this same L branch. REPL uses current-model hard labels at confidence .95 and U weight 6, six U student passes, shared cross-physical-RX style perturbations, complete-L gradient projection/capping, joint160 contrast and .25 receiver extrapolation through the existing head. There is no EMA, native neighbor/strong-view gate or new inference head. The strict-CE architecture layer instead uses the separately registered .68 channel slot and 384 L forwards, with no U/EMA/CRIC/KL; it is not an exposure-matched substitute for the multi-view core control.

PAMF-REPL was selected earlier using the already public F0 clean criterion. We therefore report a post-selection benchmark comparison, not independent blind method selection. This rerun's matrices, parameters and final-epoch rules were fixed before its new target scoring; scores do not feed back into training, tuning, seed selection, queue order or selective reruns.

F denotes the identifier and G the DURA restorer. Sup systems use G-L; PL/REPL systems use G-LU. All G ancestors match the physical source contract and train from source IQ without TX labels, F weights/outputs or target data. G-L uses L IQ only; G-LU may also use TX-hidden U. We additionally freeze both G roles for each PAMF F to separate their permissions. Sup F plus G-L is a completely L-only system; PL F plus G-L still uses U through F. No DURA training or new restored-IQ cache is performed in this rerun.

All predictions for a row's three populations, seven views and declared routes freeze before a separate CPU scorer opens truth. Accuracy and Macro-F1 use all registered classes. S6 first averages six channel views within a seed, then reports the mean and sample SD over F seeds 2026092701/02/03. Most three-F summaries condition on one fixed G; they are not three independent complete-system replicas or confidence intervals. The separate $3F \times 3G$ study provides three diagonal complete-system pairs, not nine independent repetitions.

B. Complete-Method Results and Generalization

Table III reports the main physical partitions. On F0 cross-RX, PAMF-REPL plus DURA reaches $78.29 \pm 0.97\%$ S6 accuracy, compared with $67.12 \pm 0.97\%$ for the fixed SDG D recipe. The within-seed difference is 11.16 ± 1.08 percentage points (pp), positive for all three F seeds. The corresponding REPL raw clean and raw S6 values are $86.56 \pm 0.65\%$ and $70.77 \pm 0.45\%$; after restoration clean is $86.62 \pm 0.69\%$. We retain every scene rather than infer robustness from clean alone.

The matched Sup/G-L system reaches $77.36 \pm 0.55\%$, making the complete-system difference 0.93 ± 0.97 pp. This comparison changes both F and G permissions. Holding G-LU fixed, REPL minus Sup is -0.09 ± 1.28 pp and REPL minus ordinary PL is -0.23 ± 0.95 pp. Their S6 means are nonpositive on 6/11 and 8/11 contracts, respectively. Thus the data support strong restored complete-system performance

TABLE I
MAIN PHYSICAL PARTITIONS. F1/F2 REPLACE FOUR OF THE FIVE SOURCE RX IN F0; EACH PAIR OF SOURCE SETS INTERSECTS IN ONLY ONE RECEIVER. COUNTS ARE PHYSICAL PACKETS PER VIEW.

Contract	Source RX indices	TX classes	Date shift	RX shift	Both shifts
F0	1,3,4,6,8	6	30000	126000	42000
F1	0,1,2,5,7	6	30000	126000	42000
F2	0,3,9,10,11	6	30000	126000	42000
ManyRx	Even indices 0–30	10	31279	92597	30424

TABLE II
TRAINING BUDGETS AND SOURCE-ONLY CHECKPOINT RULES. A PAMF LOGICAL EPOCH HAS 222 UPDATES. THE E20 PREFIX IS SHARED ONLY WHERE THE SAME-SEED DATA, ARCHITECTURE AND OBJECTIVE ARE IDENTICAL.

Method	Budget and optimizer	Freeze / semi-supervised objective
RIEI Full	E100; batch 64; SGD 10^{-4}	Final epoch; native three-step complete objective.
DRIFT-v2 Full	E100; batch 256; Adam 10^{-4}	Final epoch; native adversarial, center and separation terms.
IB-RFF Full	Two stages, each at most E200; batch 128; Adam 10^{-3}	Source-V minimum NLL; LR patience 5, early-stop patience 10.
SDG Full	E200; batch 32; Adam 10^{-3}	Source-V minimum NLL; native two-view and contrastive objectives.
PAMF-Sup	E50/11100; L128; native AdamW and segment schedules	Fixed final; supervised representation terms, no U/EMA.
PAMF ordinary PL	E50/11100; L128/U256	Current-model confidence 0.95, weight 1; no EMA.
PAMF-REPL	E20+10+20, cumulative E50/11100; L128/U256	Same ordinary-PL E20 prefix; selected REPL objective during E21–50.

TABLE III
S6 ACCURACY (%) UNDER MATCHING RESTORATION: G-L FOR PAMF-SUP, G-LU FOR EXTERNAL FULL+PL AND PAMF-REPL. COMPLETE RECIPES AND G PERMISSIONS DIFFER; VALUES ARE NOT SINGLE-COMPONENT CAUSAL EFFECTS. MEAN ± SAMPLE SD OVER THREE F SEEDS.

Partition / shift	RIEI	DRIFT-v2	IB-RFF	SDG	PAMF-Sup	PAMF-REPL
F0 / Date	57.87 ± 1.40	70.54 ± 0.52	90.50 ± 0.16	91.45 ± 0.21	90.99 ± 0.23	92.31 ± 0.22
F0 / RX	56.73 ± 0.91	60.75 ± 4.42	62.21 ± 2.86	67.12 ± 0.97	77.36 ± 0.55	78.29 ± 0.97
F0 / Both	57.11 ± 0.72	59.78 ± 3.53	62.84 ± 0.98	65.66 ± 0.76	75.34 ± 0.71	76.73 ± 0.88
F1 / Date	72.81 ± 0.59	75.44 ± 3.21	81.23 ± 2.87	77.95 ± 1.57	83.36 ± 0.36	84.27 ± 1.08
F1 / RX	57.39 ± 1.31	53.60 ± 3.33	55.68 ± 1.82	58.84 ± 1.15	65.84 ± 0.12	66.83 ± 0.66
F1 / Both	58.24 ± 1.72	55.47 ± 4.73	57.30 ± 2.11	58.67 ± 1.43	65.20 ± 0.34	66.91 ± 1.50
F2 / Date	65.28 ± 1.62	69.95 ± 2.40	87.61 ± 0.52	89.75 ± 0.30	87.46 ± 0.37	88.74 ± 0.21
F2 / RX	62.96 ± 1.55	57.15 ± 4.16	65.00 ± 2.36	64.34 ± 1.25	74.99 ± 0.25	74.64 ± 0.05
F2 / Both	61.53 ± 2.55	54.67 ± 4.23	64.16 ± 1.63	64.10 ± 1.91	75.25 ± 0.87	73.99 ± 0.86
ManyRx / Date	31.89 ± 6.04	46.16 ± 4.69	72.07 ± 0.42	75.00 ± 0.74	77.49 ± 0.31	77.16 ± 0.21
ManyRx / RX	28.70 ± 5.43	43.34 ± 3.64	64.99 ± 0.77	66.07 ± 0.75	71.96 ± 0.03	72.25 ± 0.39
ManyRx / Both	29.39 ± 7.82	40.92 ± 3.94	62.31 ± 0.31	66.48 ± 0.36	69.26 ± 0.77	70.15 ± 0.84

relative to these fixed external recipes, but not uniformly beneficial semi-supervision or an optimal D cell for every method.

REPL cross-RX S6 is $66.83 \pm 0.66\%$ on F1 and $74.64 \pm 0.05\%$ on F2. ManyRx cross-RX S6 Macro-F1 is $72.27 \pm 0.42\%$, using all ten classes. The label-fraction and natural source-count curves retain their negative and nonmonotonic differences. The source-count axis changes source data volume as well as RX diversity, so it does not isolate receiver count at a fixed labeling budget.

C. Functional Groups and REPL Mechanisms

The core groups are defined by function: A is labeled-source cross-RX CRIC, B is labeled-source paired symmetric KL, and C is the complete REPL U objective, including its U consistency and contrastive terms. Main rows 111/011/101/110 are accompanied by all eight combinations for marginal effects and interactions. C0 does not construct a U loader. The matched 000 CE skeleton keeps 768 L packet forwards/update.

TABLE IV
CORE REMOVAL RESULTS ON F0 CROSS-RX (%). G-LU STILL CONSUMES U WHEN F HAS C0.

Row	Raw clean	Raw S6	G-LU S6
111: Full	86.56 ± 0.65	70.77 ± 0.45	78.29 ± 0.97
011: no A	86.62 ± 0.99	70.58 ± 0.16	78.62 ± 1.08
101: no B	86.22 ± 0.66	70.35 ± 1.20	78.02 ± 0.49
110: no C	85.94 ± 0.84	71.29 ± 0.90	78.37 ± 0.43

For raw clean, full-minus-removal differences are -0.05 ± 0.83, 0.34 ± 1.17 and 0.63 ± 1.49 pp for A/B/C, respectively. The marginal B/C effects across the full factorial are 1.14 ± 0.72 and 1.60 ± 1.09 pp, while BC is -1.69 ± 0.49 pp. Marginal and full-conditional effects answer different ques-

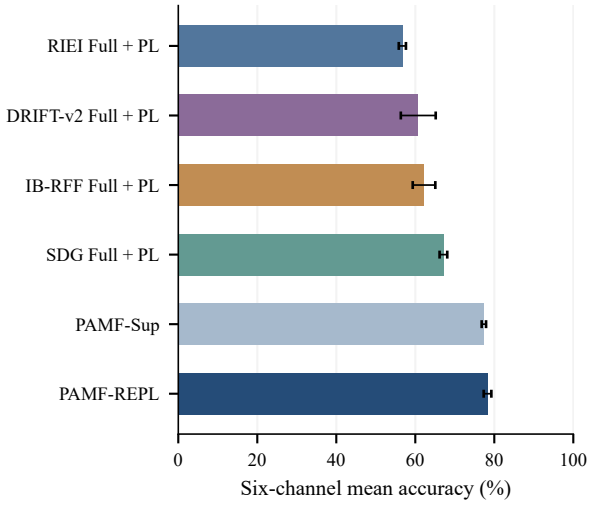


Fig. 1. Fixed complete recipes on F0 cross-RX. Error bars are three-F-seed sample SD. The external D cell is fixed in advance, not chosen by target rank.

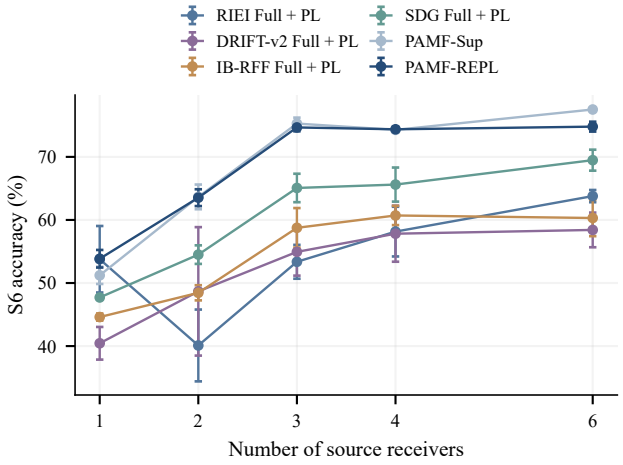


Fig. 3. Natural-budget source-count study with six fixed target RX. Only measured m=1,2,3,4,6 points are shown; F0 is not inserted as a five-source point.

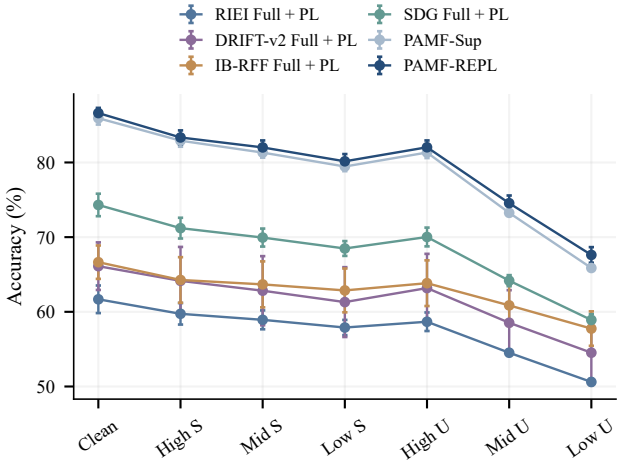


Fig. 2. All seven F0 cross-RX observations. Clean also passes through the restorer. Suburban (S) and urban (U) impairments are simulations; scene-specific weaknesses and clean changes remain visible.

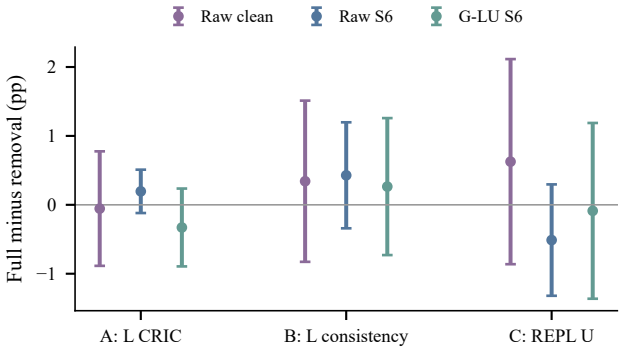


Fig. 4. Full 111 minus 011/101/110. Raw and restored effects are separate; negative and seed-dependent contributions are retained.

tions; these interactions preclude an additive interpretation. SD with three seeds is not a significance test.

The internal C ablations remove U style, gradient protection, joint-feature contrast or receiver extrapolation, each with the same E50 trajectory and head. The no-extrapolation row reuses the exact matching prior Joint160 model rather than retraining it. These are the four mechanisms of the selected REPL method; obsolete EMA/neighbor/strong-gate controls are not relabeled as REPL ablations. Full-minus-removal effects and all three seeds are supplied for every population/view/route.

Seven labeled-branch controls examine shallow-only, intermediate-only and joint-only CRIC, single stochastic forwarding, no As, As KL enabled and no raw CE. They remain diagnostic appendix rows. Single forwarding changes computation/stochastic exposure; no-raw-CE changes CE dose; the strict CE recipe changes the complete supervised objective. We do not attribute these complete-recipe differences to one isolated factor.

D. Physical Structure and Restorer Controls

The nine architectures comprise full, single time route, no time, shuffled time, no signature, no PA, no reference, fixed wrong reference and COV-X3. Each is evaluated under both full REPL and strict supervised CE, three seeds per layer. All physically matched readers use raw/G-L/G-LU; primary matching is G-LU for REPL and G-L for strict CE. The two layers differ in supervised dose and losses and are not interchangeable controls.

We reconnect seven already trained G controls to REPL for all three populations, without retraining G. Their paired F/G seeds differ from the fixed-G default summary, so the variation is not a pure-G estimate or a complete replicated copy-by-KD factorial. The three-source $3F \times 3G$ grid gives

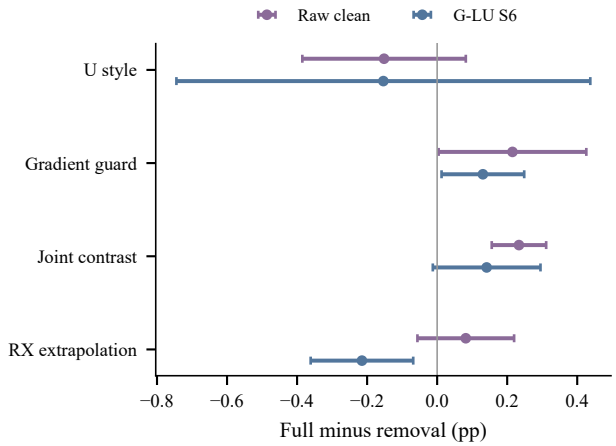


Fig. 5. Actual REPL component removals on F0 cross-RX. The same within-seed full/removal pairing is used before computing three-seed SD.

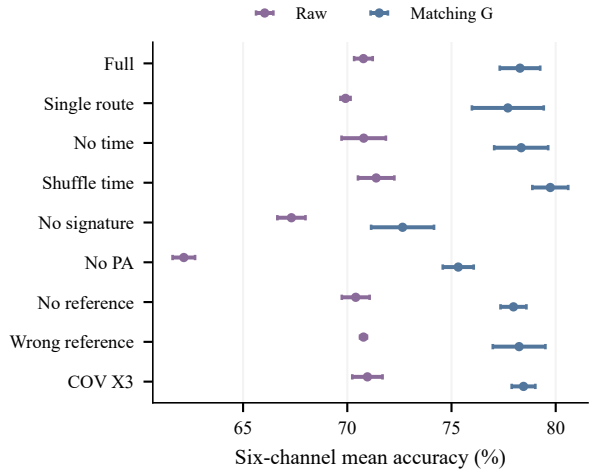


Fig. 6. Architecture sensitivity under the complete REPL recipe. Matching restoration is G-LU; all raw and alternative-G-role results remain in the tables.

diagonal S6 $74.54 \pm 1.15\%$ on cross-RX; all nine cells are supplied, and same-F raw stratified CMs agree across the extra G readouts. Only the three diagonal pairs are three complete-system training repetitions.

Source-only diagnostics evaluate the identical 720 L packets from 90 TX/RX/day strata under 11 interventions for 111 F0 models. They read no U/target and fit no diagnostic head. Default-head flips and feature/reference distances quantify sensitivity; they do not prove complete hardware disentanglement, and simulated PA distortion is not assumed to preserve TX identity.

E. Efficiency and Accounting

The updated optimized G+REPL path has B1 wall median 4.524556ms, P95 4.563896ms and CPU round-trip median 4.603213ms. Four exclusive panels cover 1175 case-rounds: REPL/strict-CE nine-architecture readers, matching Sup/ordinary-PL controls and seven existing G combinations. Batch sizes are 1/8/32/64/128; five rounds repeat timing, not training seeds. We retain 329000 timed calls and 395975 total inference invocations including warm-up, cold and transfer checks. Training-only REPL mechanisms add no inference-specific parameters. Loaded-module parameter counts and co-resident memory are retained with their measurement scope; they are not claims of active deployment MACs or isolated-model VRAM.

TABLE V
EXCLUSIVE RTX3090 B1 TIMINGS (MS), POOLED ACROSS FIVE ROUNDS. F IS EAGER; GF USES THE EXISTING OPTIMIZED GRAPH ENGINE. THESE ARE DIFFERENT EXECUTION PATHS, NOT AN ISOLATED SPEEDUP CAUSED BY G.

Case	Median	P95	CPU round trip
Sup F	7.4029	7.7351	7.6140
Sup GF	4.5629	4.5821	4.6176
PL F	7.4310	7.7949	7.3536
PL GF	4.5657	4.5854	4.6222
REPL F	7.3070	7.7823	7.2362
REPL GF	4.5246	4.5639	4.6032

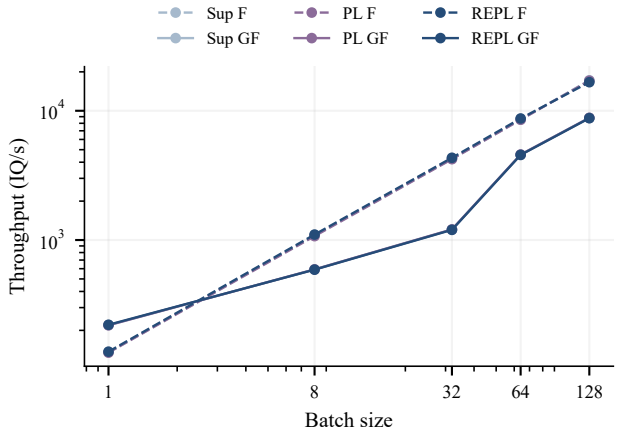


Fig. 7. Measured exclusive-device throughput. The eager-F and optimized-GF paths are explicitly separated; shared-cache classifier timing is not substituted for online G+F timing.

The 189 new F completions use 84 unique shared E20 prefixes and 8670 unique new model-epochs (1924740 updates), totaling 265.956426 shared-worker elapsed hours. The 201 logical F rows include 12 frozen F0 models and reused prefixes; summing logical cumulative costs would double-count shared stages. Each source stage, actual U acceptance and L-forward dose is retained. All C0/strict rows have zero U exposure. DURA adds zero new training and zero new restored-IQ caches in this rerun; its original training ancestry and shared-cache production are historical costs, not free computation. Known matched G costs are listed separately; the original fixed F0 G-LU ancestor timing is incompletely serialized and remains missing rather than inferred as zero.

F. Evidence Boundaries

All negative gains, zero external PL acceptance and non-monotonic source/label curves remain available. The benchmark shares WiSig acquisition hardware; synthetic residual channels do not establish in-orbit validity. Post-selection F0 clean performance, conditional F-seed variation, distinct G permissions and exposure differences constrain causal claims. The supplementary package contains the full frozen CMs/net corrections, source/epoch counters, timing samples and a pure-standard-library replay; IQ, model weights, per-sample target truth and private controller state are excluded.

APPENDIX A
COMPLETE PAMF DIAGNOSTIC ROWS

TABLE VI
LABELED-BRANCH DIAGNOSTIC RECIPES: F0 CROSS-RX ACCURACY (%). COMPLETE ROWS RETAIN EVERY SEED; EXPOSURE DIFFERENCES ARE DESCRIBED IN THE MAIN TEXT.

Recipe	Raw clean	Raw S6	G S6
Full	86.56 ± 0.65	70.77 ± 0.45	78.29 ± 0.97
As KL on	86.56 ± 1.25	71.00 ± 0.18	78.66 ± 1.11
joint only	86.50 ± 1.24	70.29 ± 0.22	78.39 ± 1.46
middle only	86.46 ± 1.42	70.81 ± 0.14	78.43 ± 1.17
shallow only	86.42 ± 1.09	71.00 ± 0.59	78.54 ± 0.99
single forward	86.62 ± 0.98	71.18 ± 0.50	78.31 ± 0.62
without As	86.26 ± 0.39	42.93 ± 5.96	78.29 ± 0.84
without raw CE	86.89 ± 1.44	71.19 ± 0.52	78.54 ± 1.12

TABLE VII
REPL ARCHITECTURE LAYER: F0 CROSS-RX ACCURACY (%). COMPLETE ROWS RETAIN EVERY SEED; EXPOSURE DIFFERENCES ARE DESCRIBED IN THE MAIN TEXT.

Recipe	Raw clean	Raw S6	G S6
Full	86.56 ± 0.65	70.77 ± 0.45	78.29 ± 0.97
COV X3	86.70 ± 0.66	70.97 ± 0.73	78.46 ± 0.57
NO PA	82.48 ± 0.13	62.16 ± 0.54	75.32 ± 0.74
NO REFERENCE	85.07 ± 0.91	70.40 ± 0.67	77.97 ± 0.62
NO SIGNATURE	82.06 ± 0.78	67.32 ± 0.67	72.65 ± 1.51
NO TIME	86.12 ± 1.77	70.79 ± 1.06	78.34 ± 1.29
SHUFFLE TIME	88.08 ± 0.85	71.39 ± 0.87	79.74 ± 0.86
SIG TIME	85.57 ± 1.41	69.91 ± 0.24	77.70 ± 1.72
WRONG REFERENCE	86.10 ± 0.72	70.78 ± 0.16	78.24 ± 1.26

TABLE VIII
STRICT-CE ARCHITECTURE LAYER: F0 CROSS-RX ACCURACY (%). COMPLETE ROWS RETAIN EVERY SEED; EXPOSURE DIFFERENCES ARE DESCRIBED IN THE MAIN TEXT.

Recipe	Raw clean	Raw S6	G S6
full	83.69 ± 1.04	71.40 ± 0.40	75.69 ± 0.53
COV X3	83.49 ± 0.98	71.68 ± 0.80	75.65 ± 0.47
NO PA	78.74 ± 0.45	64.19 ± 0.71	72.08 ± 0.59
NO REFERENCE	82.49 ± 0.88	70.82 ± 0.41	74.95 ± 0.78
NO SIGNATURE	83.36 ± 1.36	70.52 ± 0.51	75.54 ± 0.87
NO TIME	83.60 ± 1.31	70.84 ± 0.56	75.26 ± 0.37
SHUFFLE TIME	84.89 ± 1.51	72.16 ± 0.86	77.38 ± 0.69
SIG TIME	84.14 ± 0.35	70.55 ± 0.58	75.47 ± 1.20
WRONG REFERENCE	83.06 ± 0.63	71.06 ± 0.44	75.17 ± 0.61

The supplementary 27 standalone figures additionally show the complete core, internal removals, source geometry, fairness four-cell contrasts, measured architecture timing and all external Full A/B/C/D cells. They retain the same physical partitions and metric definitions as the main text.

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